

Phan Van Manh

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SUMMARY

Backend Software Engineer with over 3 years of experience building Node.js/NestJS services. Reduced a clinic-classification SQL query from ~600ms to ~40ms and redesigned an analytics pipeline over 300M+ survey records for consistent 1-2s aggregations, with a focus on database efficiency and clean, maintainable architecture.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript

Backend: Node.js, NestJS

Data: PostgreSQL, ClickHouse, OpenSearch, MongoDB, Redis; Prisma, TypeORM

Messaging & APIs: BullMQ, Kafka; REST, gRPC, WebSocket

Testing: Jest, Supertest, Cypress, Playwright

Delivery: Docker, Kubernetes, CI/CD on DigitalOcean

Observability: OpenTelemetry (traces/metrics/logs), Prometheus, Grafana, Sentry, PostHog

Process: Agile/Scrum, TDD, AI-Driven Development (AI-DD)

Principles: OOP, SOLID, DRY, KISS, YAGNI, Design Patterns

Architecture: Clean/Layered Architecture, Modular Monolith, Microservices, DDD, CQRS, Event-driven Design

EXPERIENCE

Sun Asterisk

Jul 2025 - Present

Backend Software Engineer

- Proposed and led a rebuild of the respondent-analytics module for a Japanese client: replaced OpenSearch parent-child aggregation queries on a single unpartitioned index with a ClickHouse columnar table partitioned by year, validated against a 300M+ row production copy spanning 50+ years of survey data, where it eliminated recurring query timeouts and cut typical aggregation latency to a consistent 1-2s.
- Adapted an internal AI-assisted delivery workflow (AI-DD) for a Japanese client, adding source-commit traceability so every AI-generated design document is verified against its origin before reaching developers, catching unsupported content before it causes downstream rework.

Fenik Technologies

May 2023 - Jul 2025

Backend Software Engineer

- Developed and maintained backend services for a dental-clinic discovery and review platform serving 100,000+ monthly visits.
- Designed and implemented a high-performance full-text search API using PostgreSQL's native full-text search (tsvector + GIN index) with custom relevance ranking, avoiding the cost and operational overhead of a dedicated search engine for the platform's data scale, cutting search response times by over 10x.
- Refactored the clinic classification API by partitioning the table by region and optimizing SQL queries, reducing query execution time from ~600ms to ~40ms.

PERSONAL PROJECTS

Litmatch

2024 - Present

GitHub: <https://github.com/ducmanh001/litmatch> | Demo: <https://litmatch-web.manhphan110801.workers.dev>

- Solo-designed and built Litmatch, an anonymous social platform (matching, chat, feed, mini-games, diamond/VIP economy), as a NestJS modular-monolith monorepo (Socket.IO, LiveKit SFU, Next.js) with 20+ modules.
- Solved the dual-write problem for diamond/VIP transactions with a transactional outbox: business writes commit atomically with an outbox row, then a relay job (Postgres FOR UPDATE SKIP LOCKED, safe across concurrent instances) publishes to Kafka and marks it sent - so pending events remain retryable after failures instead of being silently lost.
- Built the matching engine as a sharded priority queue with a trust-score penalty and an idempotent diamond-priced boost, plus a two-sided confirm locking session and ticket in fixed order to avoid deadlock.

Mini Shop API

2025 - Present

GitHub: <https://github.com/ducmanh001/mini-shop-api> | Demo: <https://mini-shop-api-production-e25a.up.railway.app/api-docs>

- Solo-built a NestJS/TypeScript e-commerce backend covering auth, products, cart, checkout, orders, reviews, and admin management, with PostgreSQL/TypeORM migrations and Bull-queued email notifications, deployed to Railway with Docker and CI; checkout uses ordered row locks and idempotency keys to reduce deadlock risk and prevent duplicate orders on retry.

EDUCATION

FPT Polytechnic

Web Programming

Sep 2020 - Feb 2023